

Letter to Millson

Dear Millson,

Nori brought me a copy of your paper with Goldman on deformations of flat bundles. It seems to me that it contains a number of mistakes, but also that, when corrected, it gives the strong results you ask for $p \leq 7$. I begin with the latter.

The philosophy, which I had not realized before reading your paper, seems the following: in characteristic 0, a deformation problem is controlled by a differential graded Lie algebra (d.g.l.a.), with quasi-isomorphic DG Lie algebras giving the same deformation theory. If the DG Lie algebra controlling a problem is “formal”, i.e. quasi-isomorphic to $\oplus H^i$, then the versal (formal) deformation space is that of $\oplus H^i$, i.e. is the completion at 0 of the subscheme of H^1 defined by the equations $[u, u] = 0$.

If one believes in this, what you do becomes transparent:

Deformation problem: G is a given reductive group, and one wants to deform a given G -torsor P — i.e. a $G(\mathbb{C})$ -torsor, or, equivalently, a $G(\mathbb{C})$ -torsor, with integrable connection ∇ .

Controlling DG algebra: (one is over $\mathbb{C}$):

$$\Omega^{**}(\mathrm{Lie} G^P)$$

If P can be reduced to a maximal compact $K \subset G$ (i.e. to a compact form of G), Hodge theory shows that the controlling

algebra is “formal”: one has quasi-isomorphisms of DG Lie algebras

$$\Omega^{**}(\mathrm{Lie} G^P) \longleftrightarrow \mathrm{Ker}(d') \longrightarrow \frac{\mathrm{Ker}(d')}{\Im(d'')} \quad (1)$$

and $\mathrm{Ker}(d')/\mathrm{Im}(d'')$ has $d = 0$. To check the above, it is easiest to use the principle of the 2 types in the form that the double complex $\Omega^{**}(\mathrm{Lie} G^P)$ is (forgetting $[\cdot, \cdot]$) a sum of

- a) a double complex with $d' = d'' = 0$ (“harmonic part”)
- b) squares of isomorphisms: (to shewline)

$$\begin{array}{ccc} * & \xrightarrow{d'} & * \\ d'' \uparrow & & \uparrow d'' \\ * & \xrightarrow{d'} & * \end{array}$$

As far as formal deformations are concerned, this concludes the story. It is not clear to me how to use Artin in the analytic case [sol. of anal. eqn.] to get the same for analytic-versal. The algebraic case is OK, but Artin is more powerful there.

Here is how to give a meaning to the philosophy (and check it):

Construction: How to attach to a DG Lie algebra L^* (over a chosen field k) a fibred category over the category of $\mathrm{Sp}(A)$, A local k -algebras of finite dimension with residue field k .

Object over $\mathrm{Sp}(A)$: $[k = A/m]$

$$\omega \in L^1 \otimes m, \quad \text{such that} \quad d\omega + \frac{1}{2}[\omega, \omega] = 0$$

Map: The Lie algebra L^0 gives rise to a formal group $\widehat{L}^0$; the A -points of this formal group are the $L^0 \otimes m$, viewed as a nilpotent Lie algebra, and equipped with Campbell–Hausdorff group law.

Notation: $\widehat{L}^0(A)$.

The formal group $\widehat{\underline{L}}^0$ acts on L^1 (thanks to $[\cdot, \cdot]: L^0 \otimes L^1 \rightarrow L^1$), hence $\widehat{\underline{L}}^0(A)$ on $L^1 \otimes A$. One has also a map

$$“g^{-1}dg”: \widehat{\underline{L}}^0 \longrightarrow L^1 \quad : \quad \exp(X) \longmapsto \frac{1 - \exp(-\operatorname{ad} X)}{\operatorname{ad} X} [dX]$$

i.e.

$$\widehat{\underline{L}}^0(A) \longrightarrow L^1 \otimes m$$

One now defines a map from $\omega \in L^1 \otimes m$ to $\omega' \in L^1 \otimes m$ as being $g \in \widehat{\underline{L}}^0(A)$ such that

$$\omega' = g dg^{-1} + g(\omega) \tag{2}$$

One has to check that $d\omega' + \frac{1}{2}[\omega', \omega'] = g(d\omega + \frac{1}{2}[\omega, \omega])$ when ω and ω' are related by (2).

To be checked: for a square zero ideal I in A , and an object ω on A/I , there is an obstruction in $H^2(L \otimes I)$ to extend ω to A ; if the obstruction vanishes, isomorphism classes of extensions form a principal homogeneous space under $H^1(L \otimes I)$; automorphisms of any extension (trivial on A/I) are $H^0(L^* \otimes I)$. This is functorial in L^* , hence the fact that quasi-isomorphisms induce equivalences of fibred categories.

Application here: Let X be an analytic variety and (V, ∇) be a (holomorphic) vector bundle with connection. The deformation problem is

$$A \longmapsto \text{category of vector bundles with connection in the } X\text{-direction,}$$

on $X \otimes \operatorname{Sp}(A)$, deforming (V, ∇) .

It is given by the previous construction applied to $\Omega^{**}(\operatorname{End} V)$.

Rmq: In your application, because you have the explicit quasi-isomorphism (1), you have an explicit equivalence of fibred categories between

- * deformations of (V, ∇) as above
- * category with objects over $\mathrm{Sp}(A)$: $\mathrm{Sp}(A) \longrightarrow$ subscheme of H_L^1 with $[u, u] = 0$
and arrows: $g \in (\underline{H}_L^0)^\wedge(A)$ from u to $g(u)$

The functor T is such: a cohomology class $u \in H_L^1 \otimes m$, with $\varepsilon_u: \mathrm{Sp}(A) \rightarrow$ scheme H_L^1 , can be uniquely written as $\omega \in L^1 \otimes m$ with $d'\omega = d''\omega = 0$. This ω can be corrected by a $d''\beta$ to have zero curvature, if $[u, u] = 0$. The corrected $\tilde{\omega}$ is unique mod $\exp(\mathrm{Ker} d' \otimes m)$.

Rmq: (1) is a filtered quasi-isomorphism, for the Hodge filtration. This should translate into saying that the pair of spaces

$$(\text{deformations of } (V, \nabla)) \supset (\text{deformations of } \nabla \text{ on a fixed } V)$$

is isomorphic to the pair

$$(H^1 \supset F^0 H^1) \cap \mathrm{Grass}[u, u] = 0.$$

II. Comments on your paper

p 2 after cor. 3: no [take $G = U_1$]

p 6 l. 18 holomorphic? : I guess you want to start with η such that $d'\eta = d''\eta = 0$

p 7 l. 4 : why closed $\Rightarrow$ (2,0)-component closed?

III. Generalization

Hodge theory is available for variations of Hodge structures (complex variations, on compact Kähler manifolds). The useful bigrading of $\Omega^{**}(\text{End } V)$ is obtained by mixing that of Ω^* and that of $\text{End } V$. It is useful now that (1) is a filtered quasi-isomorphism. One gets:

- a) The same result as before on deformations of a representation of π_1 given by a complex variation.
- b) Looking at the deformation theory controlled by $F^0(\dots)$: same result for the versal moduli of a filtered bundle with connection coming from a complex variation. One insists that the deformed connection must continue to satisfy “transversality”:

$$\nabla F^i \subset \Omega^1 \otimes F^{i-1}.$$

Yours sincerely,

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